

Reporte de Análisis Físico

Validación MC y Asimetría UMD/SD

Configuración de la Simulación

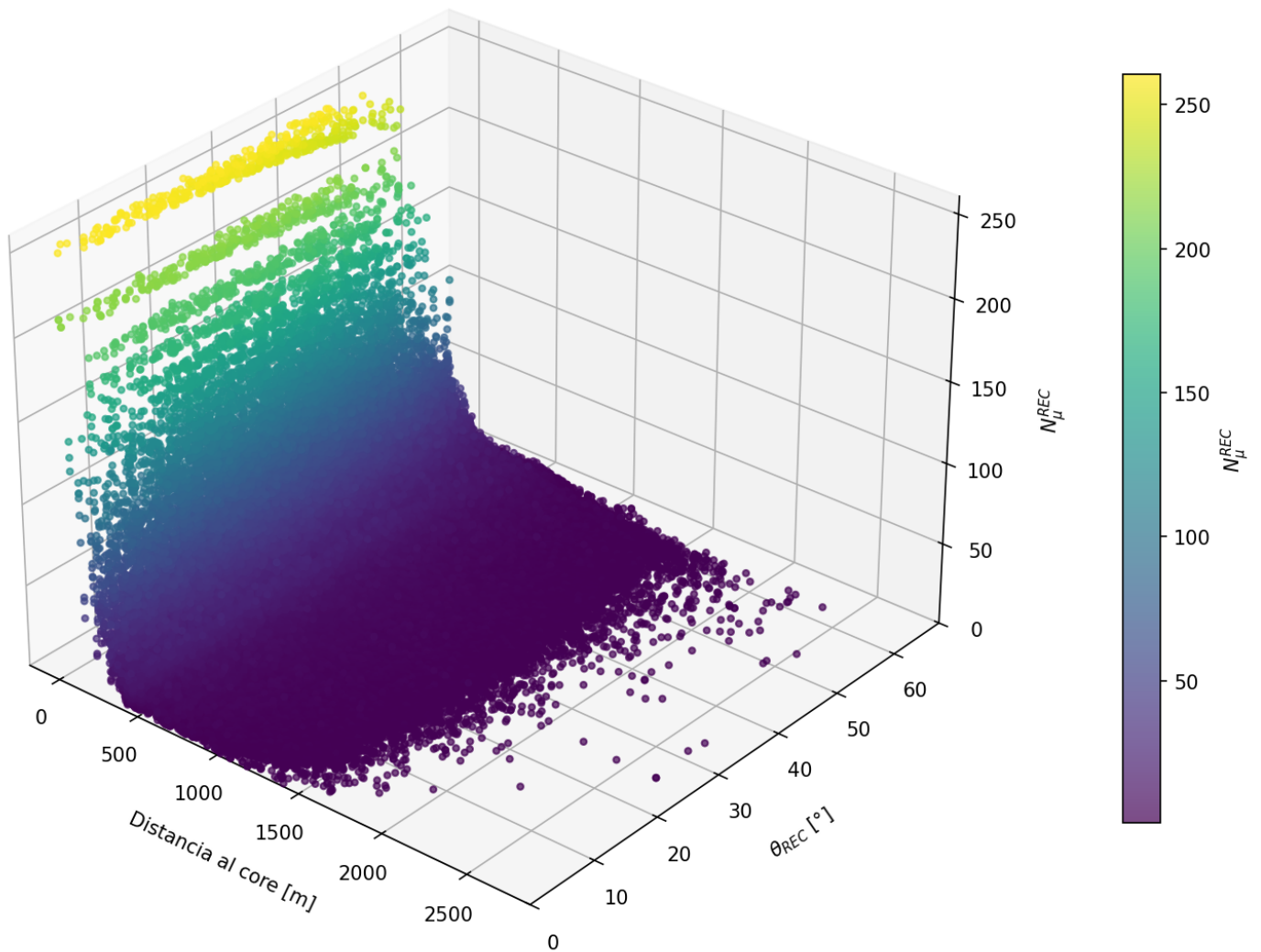
Parámetro	Valor
Modelo Hadrónico	SIBYLL 2.3d
Primario	Helio
Rango Energía	17.5 - 18.0 log(eV)
Carpeta	
Eventos Totales	4,986
Módulos Totales	1,623,057

1. Limpieza de Datos

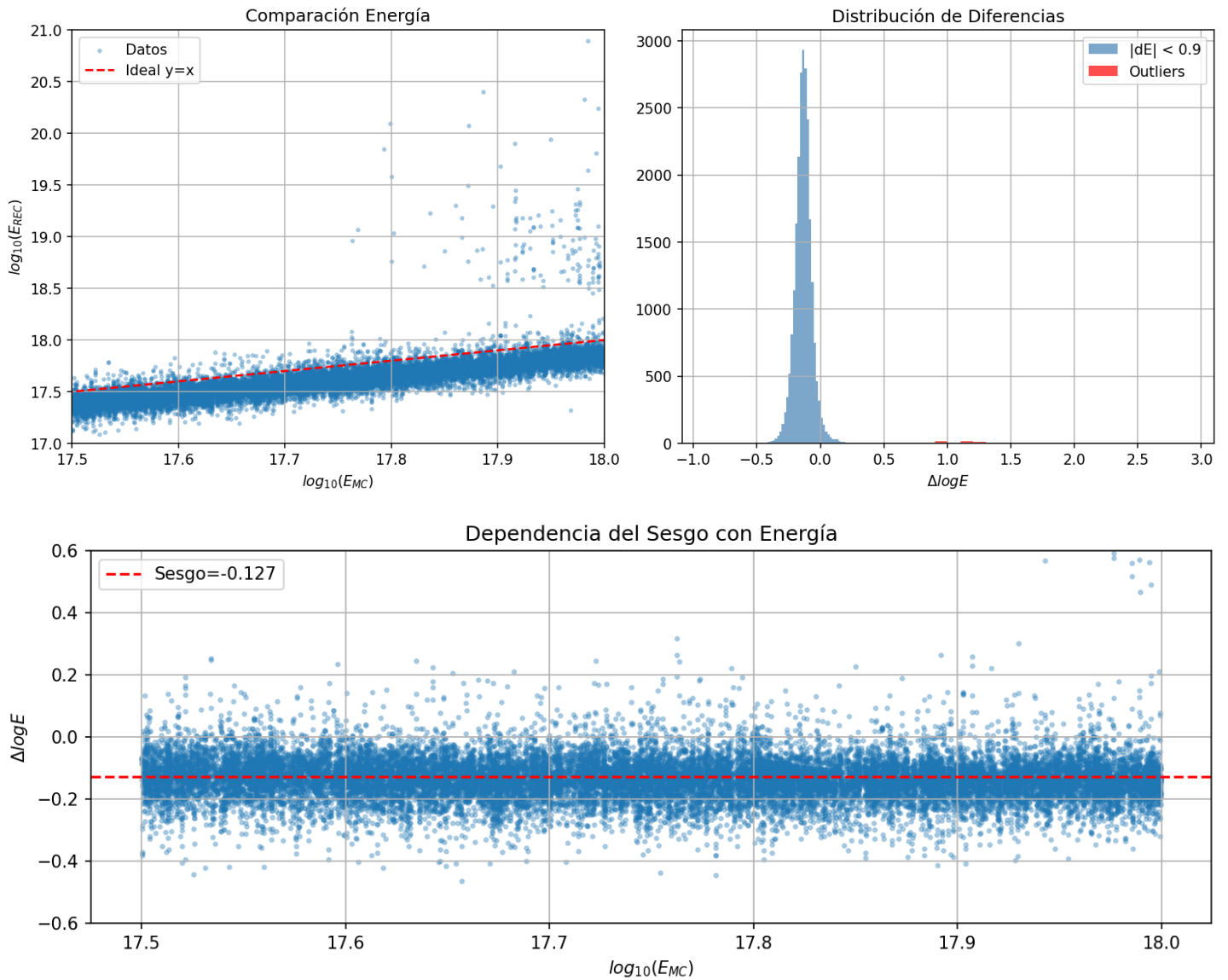
REPORTE DE LIMPIEZA

Métrica	Valor
Filas Iniciales	1623057
Filas Eliminadas	59040 (3.64%)
Filas Finales	1564017
Causa Principal	NaN en logE_REC (59040)

2. Visualización General

N_{μ}^{REC} vs distancia y θ 

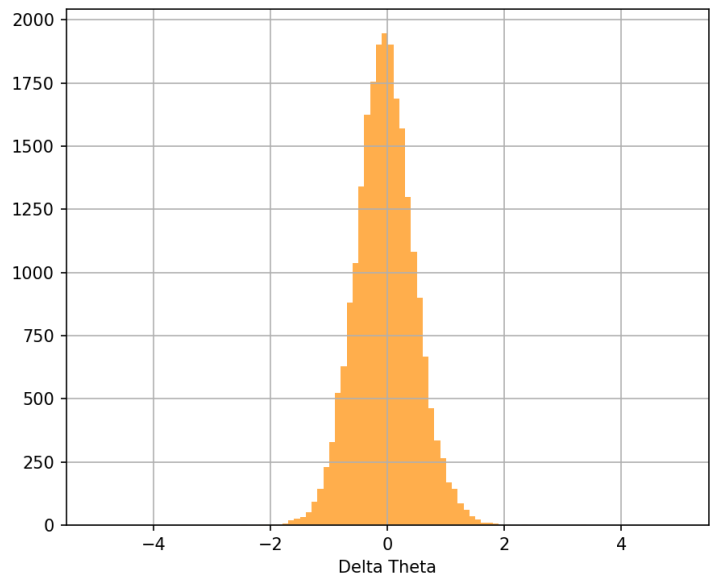
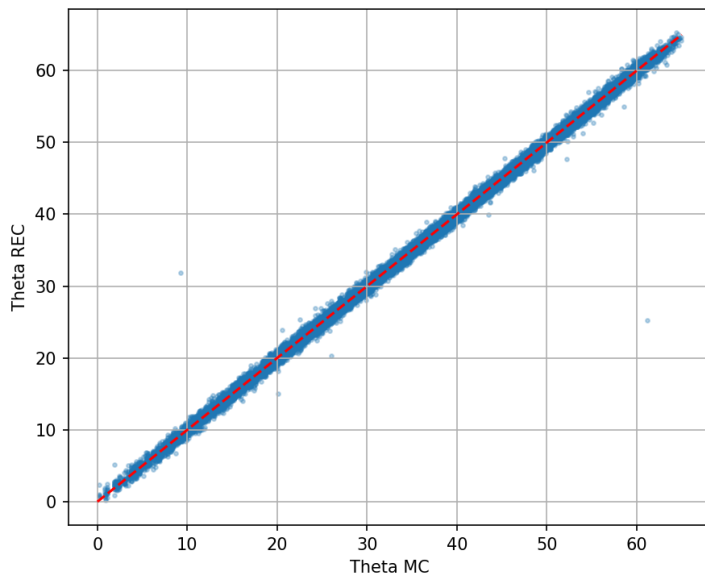
3. Resolución de Energía



Sesgo (Mean): -0.1270
 Resolución (Std): 0.1179
 Outliers (> 0.9): 75 (0.32%)

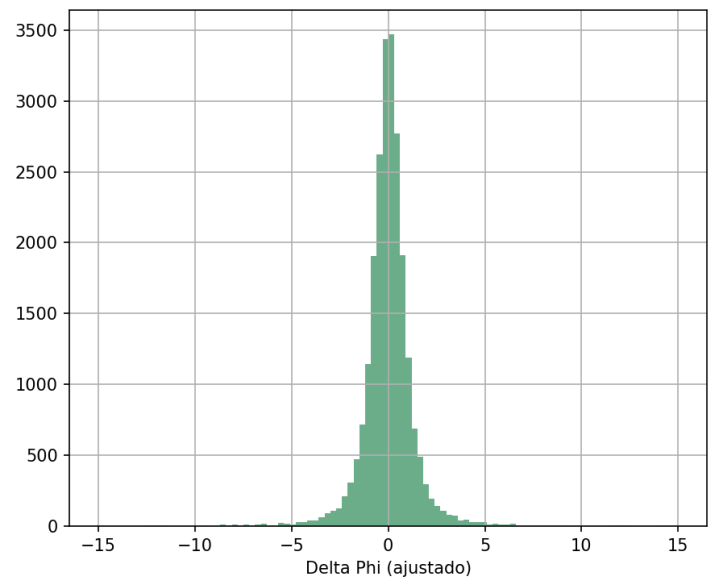
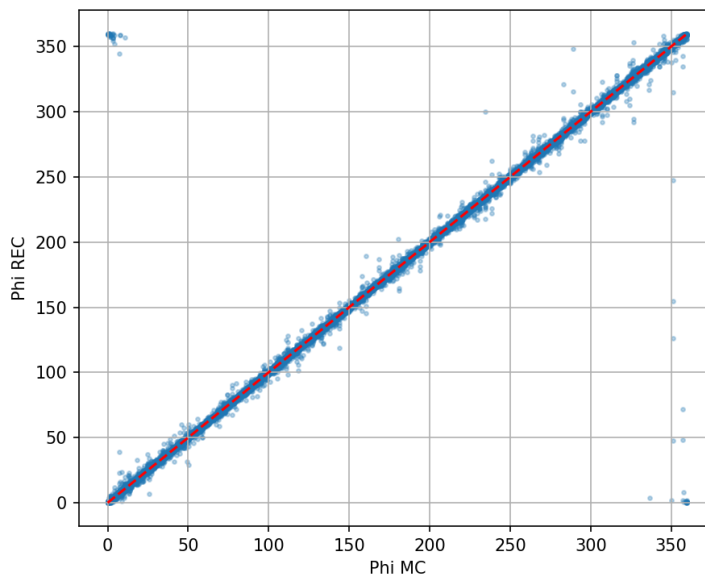
4. Resolución Angular

Theta (Cenital)



Sesgo: -0.0413 deg
Resolución: 0.5828 deg
Outliers ($> 10.0^\circ$): 2 (0.01%)

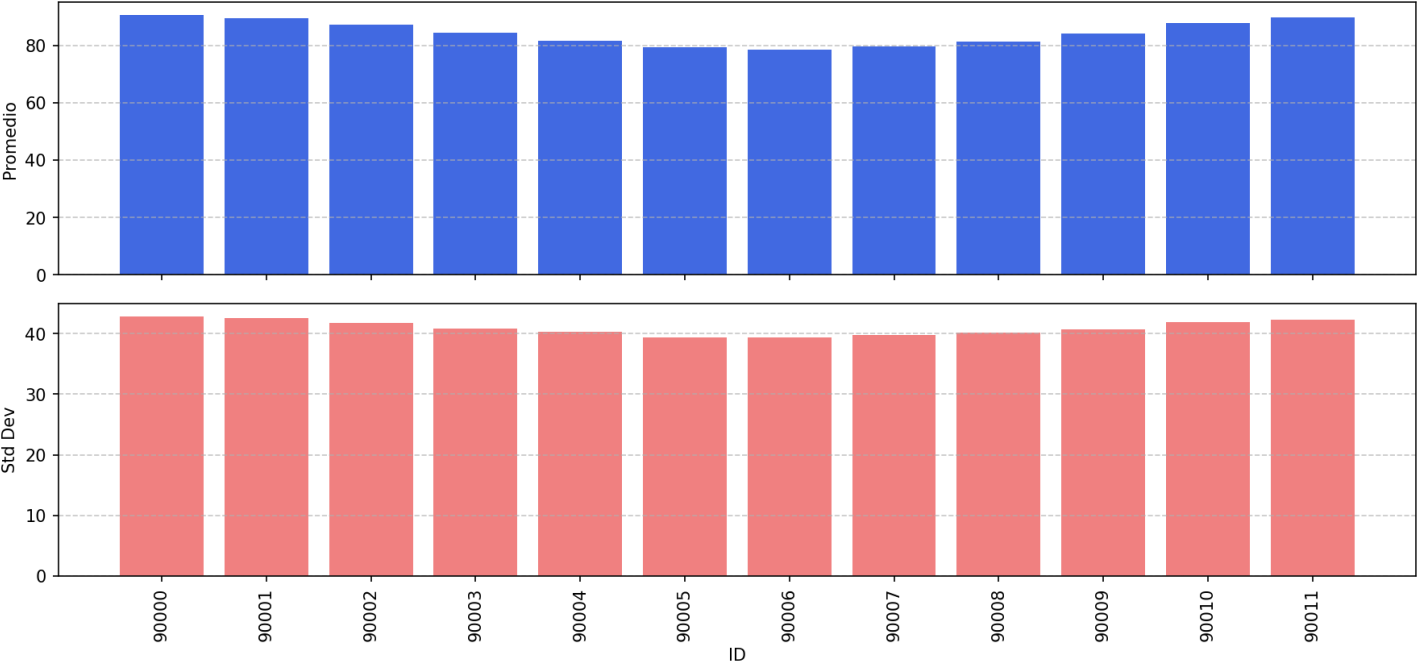
Phi (Azimutal)



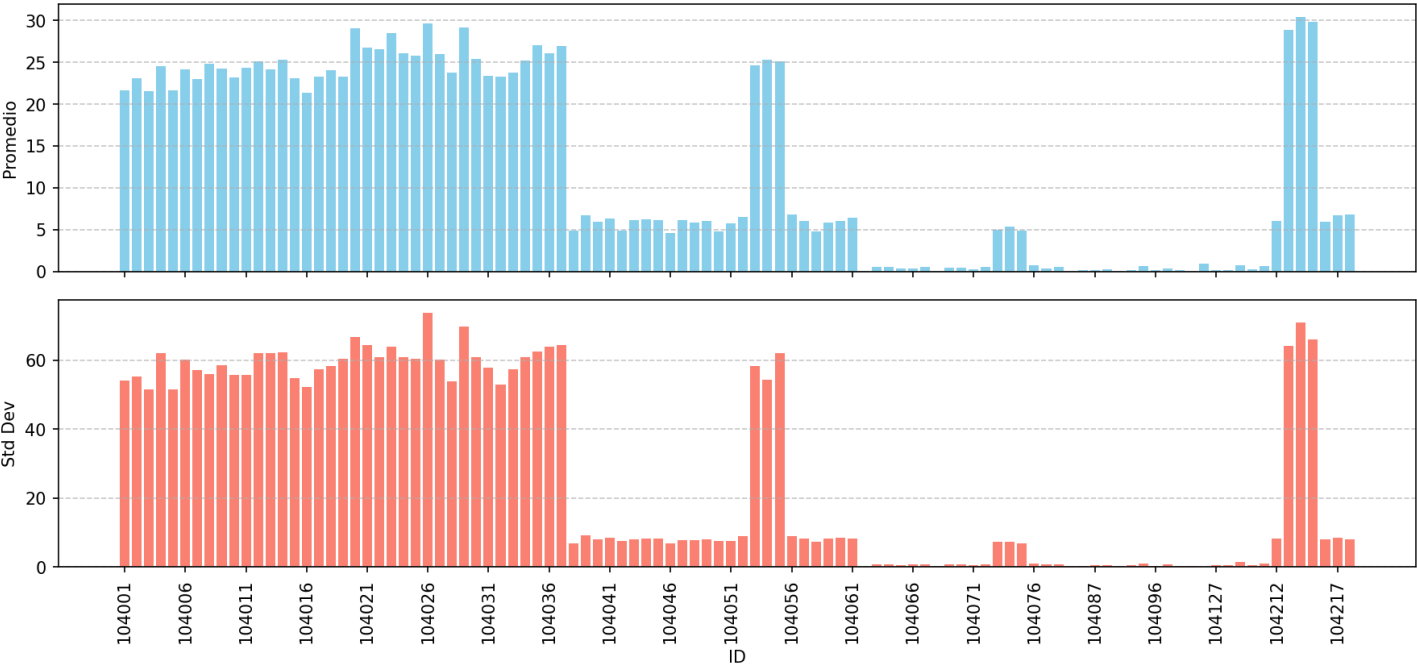
Sesgo (Median): 0.0105 deg
Resolución (MAD): 0.5599 deg
Outliers ($> 15.0^\circ$): 43 (0.18%)

5. Uniformidad por Counter

Uniformidad: Anillo Denso (90k)

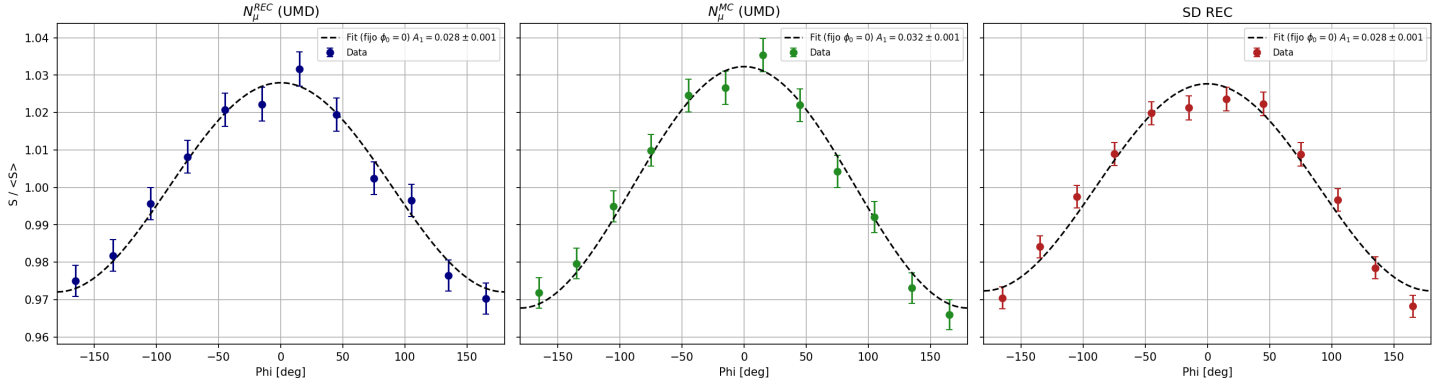


Uniformidad: Infill (104k)

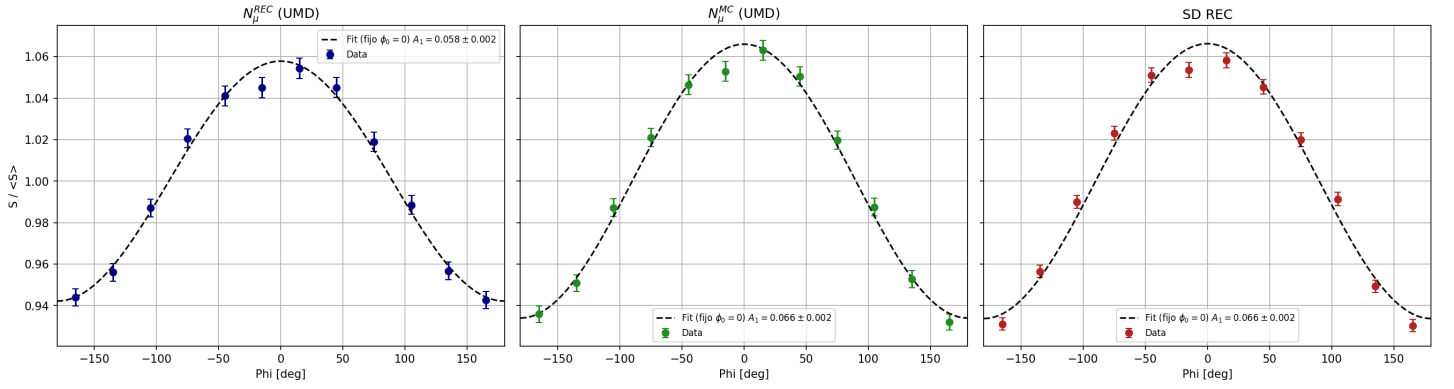


6. Asimetría Azimutal

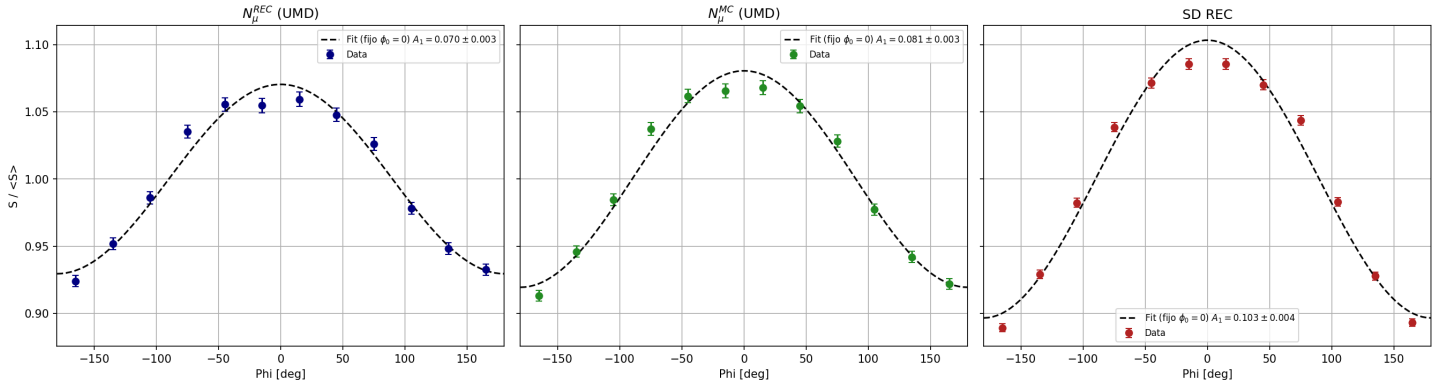
Bin 0: Theta 0.2 - 21.7 deg



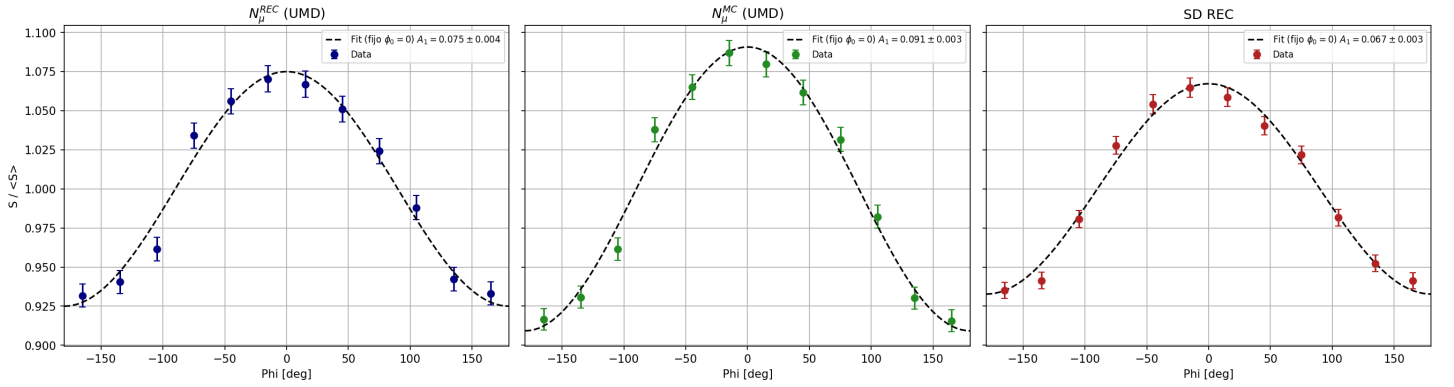
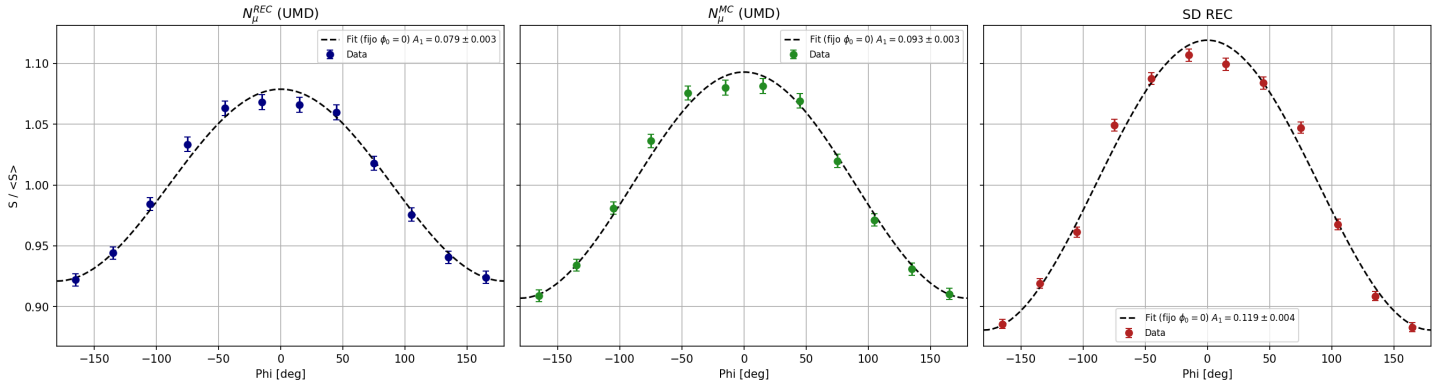
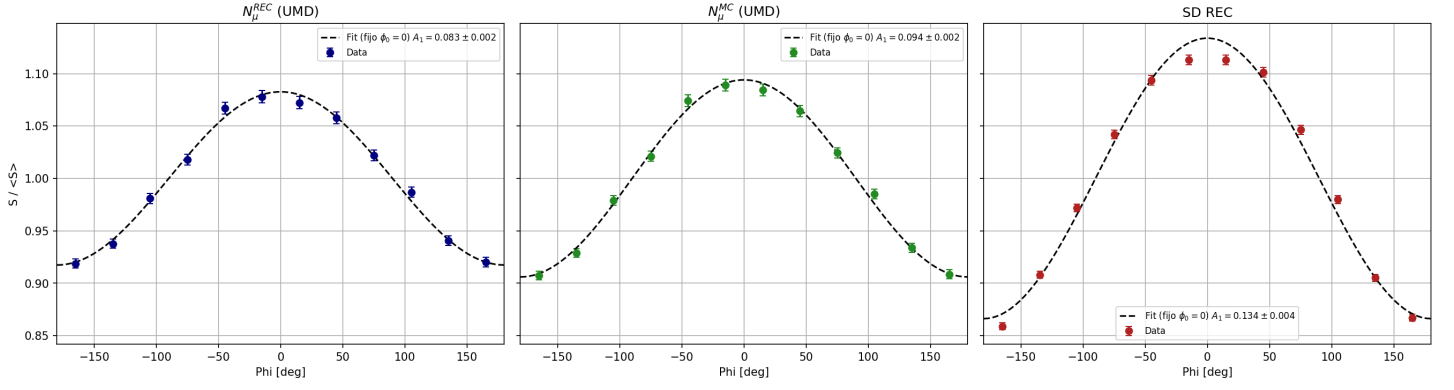
Bin 1: Theta 21.7 - 31.6 deg



Bin 2: Theta 31.6 - 39.9 deg



Bin 3: Theta 39.9 - 47.7 deg



7. Resumen de Ajustes (Comparación)

Valores de Amplitud A1

theta_range	A1_UMD_REC	A1_UMD_MC	A1_SD_REC
0.2-21.7	0.0280	0.0323	0.0277
21.7-31.6	0.0579	0.0660	0.0663
31.6-39.9	0.0705	0.0806	0.1034
39.9-47.7	0.0826	0.0940	0.1338
47.7-55.8	0.0789	0.0930	0.1192
55.8-65.0	0.0750	0.0908	0.0673

